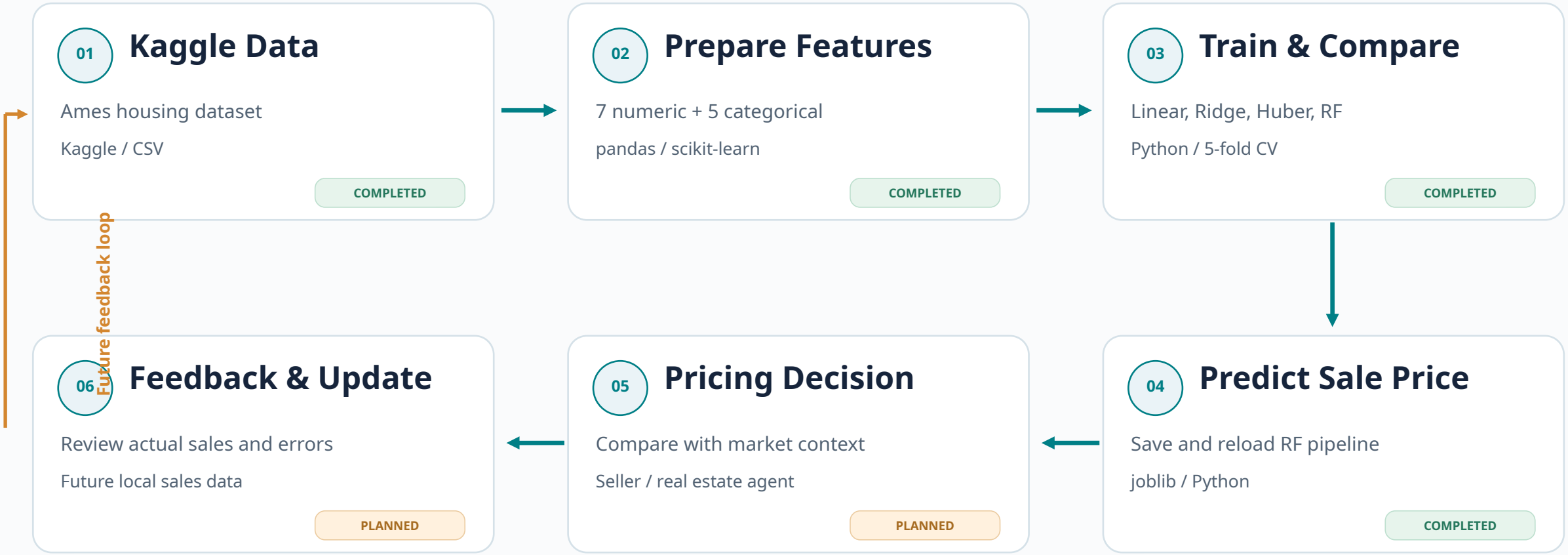
House Prices | Data-to-Action Pipeline

MSBA 265 Final Project • Current model workflow and planned business use



From historical sales data to a pricing reference — not a formal appraisal.

Model Performance & Business Value

Results reproduced after a fresh GitHub clone and virtual-environment setup

$29,971	$18,112	0.8829
RMSE	MAE	R^2

What the model does

Uses 12 house features to estimate a sale price. The saved Random Forest pipeline includes categorical encoding, and its predictions can be reproduced in a new environment.

How someone could use it

A seller or real estate agent could compare the estimate with local listings and recent sales. They should still consider condition, location, and current market changes.

Important limitation The data comes from Ames, Iowa. This is a research prototype, not a current-market appraisal tool. Business use and live feedback are planned, not deployed.

Evaluation: 292 held-out records • 1,168 training records • Model: Random Forest with 12 input features